Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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Project Progress

Create Records in the Sheet

Create Import Set Table (Staging Table)

Create Transform Map

Field Mapping (Mapping Assist)

Use Dot Walking for Relationship Mapping

Transform the Data

Display Relationship Data

Result

Conclusion

Introduction

- In ServiceNow, importing external data (like Excel/CSV) into tables can be automated using Import Sets.
- To establish relationships between tables (like User → Department), we use Dot Walking.

  - Reduce manual data entry
  - Maintain data consistency
  - Automatically link related records

Create Records in the Sheet

- Prepare your data in Excel/CSV.
- Column names are clear
- No duplicate or invalid values

Create Import Set Table (Staging Table)

- Go to: System Import Sets → Load Data
- Upload your Excel file
- Create a staging table (temporary table)

Create Transform Map

- Navigate to: System Import Sets → Transform Maps
- Create a new Transform Map:
- Source table → Import Set Table
- Target table → e.g., User [sys_user]

Design Philosophy

Data deserves clarity and respect.

Field Mapping (Mapping Assist)

Use Auto Map Matching Fields or manually map:
Example:

- Name → Name
- Email → Email
- Department → Department

Use Dot Walking for Relationship Mapping

Dot walking allows you to access related table fields.

[Department → sys_user.department.name]

- Department is stored in another table (cmn_department)
- Instead of storing plain text, you link records

[Department → Department.name]

Transform the Data

- Click Transform
- Data moves from staging → target table

Behind the scenes:

- Relationships are created
- Data is validated
- Errors are logged

Display Relationship Data

Now you can use Dot Walking in forms or lists:

Example:

[Assigned to → Department → Name]

- User's department dynamically

Result

Users imported successfully
Departments linked correctly
No manual relationship creation needed

Conclusion

Using Import Sets + Transform Maps + Dot Walking in ServiceNow enables:

- Automated data import
- Accurate relationship mapping
- Scalable data handling
- Reduced human errors.

Process Overview

What story does the data want to tell?

Understand the Objective

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking is a method of efficiently importing external data into a system

Review and Audit the Data

Review and Audit of Data is the process of verifying imported data for accuracy, completeness, and correct mapping to ensure proper transformation and relationship handling in the system.

Sketch Information Flow

This automation reduces manual effort, ensures data accuracy, and establishes relationships between records (like linking users to departments).

Build Visual Hierarchy

This approach enables smart data handling, where not only data is imported but also relationships are intelligently built, making the system more dynamic and efficient.